SAPTAK BANERJEE

Software Developer · Data Analyst · Quantitative Researcher

BCA First Division (IGNOU) · **2 shipped open-source plugins** · **HSMM-based trading engine** · Full-stack + quant research

📍 Kolkata, WB · 📞 8240323276 · 🐙 **github.com/nilarkian**

CORE SKILLS

Languages: Python · TypeScript · JavaScript · PHP · SQL · C · Java · Lisp
Quantitative: NumPy · Pandas · QuantLib · Backtesting · Regime Modeling · Signal Design · HSMM
Web & DB: PHP · MySQL · Chart.js · CSS · HTML · XAMPP · REST
Tools: Vim · Git · Obsidian · Anaconda · Jupyter · Linux (WSL)
Analytics: Time-series Analysis · Feature Engineering · Statistical Modeling · Behavioral Analytics
Design: UX/UI — Google/Coursera Certified · CSS Component Architecture

PROJECTS

Obsidian Plugins — pill-view & Hill-Chart-Interactive Open Source · github.com/nilarkian *2025 – 2026*

- Shipped **2 production plugins** in TypeScript, JavaScript & CSS — each a fully custom Bases view with real-time UI interactions.
- Built **pill-view**: pill-tab navigation component switching active Bases views on click — **zero page reload**.
- Built **Hill-Chart-Interactive**: draggable SVG hill-chart synced bidirectionally to note frontmatter via property watchers — live project-progress tracking inside Obsidian.
- Learned **TypeScript + Obsidian plugin API** from scratch to ship both; both independently installable and publicly documented.

Kinetic Regime Synchronization — Quantitative Trading Engine Independent Research *2026 – Ongoing*

- Architected **Hidden Semi-Markov Model (HSMM)** with Poisson-Driven Jump-Duration enforcing a **25-round minimum persistence constraint** — regimes as latent states, not post-hoc filters.
- Engineered **zero-memory Relative Price Flux observable (Δ+ / |Δ−|)** with **≥1.6 Pareto-tail threshold** — eliminating estimator inertia inherent in moving-average systems.
- Built **Python/Pandas/NumPy walk-forward backtesting engine** covering Online Viterbi Decoding, regime detection, adaptive threshold tuning & payoff stability diagnostics.
- Based on **Tit-for-Tat (TFT) algorithm from game theory** and payoff matrix analysis (CC/DC/DD/CD) to model cooperation-defection dynamics in intraday price action.

Financial Management System — Peer Performance Index Academic Project *2025*

- Built **full-stack financial analytics platform** — PHP, MySQL, JavaScript, Chart.js — covering auth, CRUD, reporting modules & interactive KPI dashboard.
- Designed **Peer Performance Index**: custom scoring metric benchmarking individual financial behavior against cohort averages — finance + analytics + software in one deployable system.
- Delivered **end-to-end system architecture** — database schema design through data visualization layer — translating business problems into production software.

Currency Market Volatility Model Final Year Project*Dec 2021 – 2022*

- Built **multi-asset FX volatility model** using Python and QuantLib — translating academic pricing theory into a working quantitative tool.
- Modeled **price dynamics across currency pairs** — enabling quantitative risk assessment of FX positions.

EXPERIENCE

Content Writer LBB Kolkata *Sep 2019 – Jul 2020*

- Identified **high-converting content patterns** by analyzing which editorial topics drove reader spending — early applied behavioral analytics.
- Published editorial analysis on Kolkata's F&B sector — covering price competition, market saturation & venue differentiation.

Policy Researcher LBB Kolkata*Jan 2019 – Sep 2019*

- Mapped **seasonal transport shifts to consumer spending behavior** across dining, retail & entertainment sectors in Kolkata.

Content Writer Nitch (Remote) *Jun 2020 – Dec 2020*

- Authored **6 months of long-form editorial content** tracking structural evolution of the global music industry — streaming economics & independent artist growth.

EDUCATION

BCA — Bachelor of Computer Applications IGNOU (Indira Gandhi National Open University)*Jul 2022 – Dec 2025*

First Division · **67.61%** · 1893 / 2800 · 99 Credits

Coursework: Data Structures · Algorithms · OOP (Java/C++) · DBMS · Systems Analysis · Web Programming · Computer Networks · Software Engineering · Statistical Techniques

ICSE — Science National Gems Higher Secondary School*2018*

CERTIFICATIONS

Google UX/UI Design — Coursera · Certified

Currently Pursuing — Microsoft & LinkedIn Learning:

- Microsoft Azure **AI Essentials** Professional Certificate
- Career Essentials in **Data Analysis** — Microsoft & LinkedIn
- Career Essentials in **System Administration** — Microsoft & LinkedIn

LEADERSHIP & COMMUNITY

- Coordinated events and managed operations at **MUN Circle, Kolkata**.
- Conducted **3 awareness campaigns** as Youth Organizer — Calcutta Rotaract Club, Art of Life.